Trigger-Adaptive Sampling for Rare-Event Capture in Low-Cost Power Quality Monitoring on Microcontrollers

Abstract

Low-cost power-quality monitoring systems deployed in unstable-grid environments must preserve short duration voltage disturbances while operating under stringent memory and energy constraints. Fixed-rate acquisition provides temporal coverage but continuously pays the resource cost required to resolve the fastest expected event. Existing adaptive sampling approaches are commonly designed for slowly varying signals and do not explicitly address the asymmetric cost of missing a rare transient. This paper presents the Trigger Adaptive Sampling Algorithm (TASA), a two-state sampling strategy designed for rare-event capture on resource-constrained microcontrollers. TASA operates at a low baseline rate during stable conditions and transitions deterministically to a full capture rate when a lightweight trigger detector identifies a potential disturbance, while retaining pre-trigger context in a circular buffer. The implementation is written in C and evaluated through firmware simulation on an STM32F103 target in Wokwi using a labeled synthetic power-quality signal containing a voltage sag and voltage swell. With r_b = 50Hz, r_f = 200Hz, N_{pre} = 32, N_{post} = 200, and hysteresis H = 10, the full TASA finite-state machine captures all 80 ground-truth event samples, yielding 100% event-sample recall and 71.4% precision. The compiler-verified static state footprint is 588 bytes. TASA uses 888 baseline samples and 112 triggered samples, corresponding to a time-weighted average sampling rate of 54.59 Hz and a 72.7% reduction relative to the 200-Hz fixed-rate reference. A faithful C re-implementation of EcoTrack is also evaluated as a deliberately cross-domain comparison. Its internal rate parameter collapses to zero throughout the sag and is zero for 947 of 1000 evaluated ticks; because the port does not define EcoTrack's integer parameter k directly as a frequency, it is not converted to Hz. The results support the use of explicit trigger-based state transitions for rare, fast power-quality event capture, while also showing that the present evidence remains simulation-based and requires validation on physical hardware and recorded grid disturbances.

Keywords: adaptive sampling, power quality monitoring, transient detection, event-triggered sampling, embedded systems, microcontrollers, rare-event capture

INTRODUCTION

Power-quality (PQ) disturbances such as voltage sags, voltage swells, interruptions, transients, and waveform distortions can affect electrical equipment and power-electronic systems. IEEE Std. 1159 provides terminology and recommended practices for monitoring electric power quality, while IEC 61000- 4-30 specifies measurement methods for a range of PQ parameters [1], [2]. The measurement problem becomes particularly difficult when monitoring must be distributed widely using inexpensive embedded hardware. Nigeria provides a relevant deployment context because distributed power-quality monitoring can complement existing measurement infrastructure by providing event-level observations at locations that are otherwise difficult to instrument densely. Prior work has demonstrated embedded and low-cost approaches to power-quality monitoring, including dedicated sensor nodes and microcontroller-based analyzers [3]-[5]. Such systems motivate the need for sampling strategies that preserve short-duration disturbances without imposing the cost of continuous high-rate acquisition.

Commercial PQ analyzers provide substantially more measurement capability than a small microcontroller logger, but their cost and deployment complexity can make dense monitoring difficult [4], [6]. A low-cost logger therefore faces a basic tradeoff: continuously sampling at a rate high enough to resolve the fastest expected event consumes memory, processing activity, and potentially energy even during long stable intervals [6]. Conversely, reducing the rate without an explicit event-capture mechanism can cause a short disturbance to occur entirely between samples [6],[7].

Adaptive sampling reduces acquisition activity by changing the sampling rate according to signal or resource conditions. It has been studied extensively in sensor networks, environmental monitoring, energy-harvesting systems, and IoT data streams [8]-[12]. EcoTrack, for example, uses a finite-state machine with additive increase and multiplicative decrease to adapt sampling activity according to battery state of charge in an energy-constrained IoT device [13]. Power-quality transient capture has a different objective. The monitored signal may remain near nominal for long periods and then change substantially over a short interval [14]. More importantly, the cost function is asymmetric: an unnecessary sample during a stable interval has a small cost, whereas failing to acquire the samples associated with a short disturbance can make the event unavailable for subsequent analysis [7], [6].

This paper therefore studies whether a sampling strategy explicitly organized around event detection is more appropriate for this problem than continuous rate adaptation. The central design idea is simple: maintain a low-rate observation and a short rolling context during stable operation, then switch deterministically to full-rate capture when a trigger is detected.

The study asks: Can a lightweight two-state trigger-based sampling strategy reduce average sampling activity on a resource-constrained microcontroller while preserving complete capture of rare power-quality events in the evaluated scenario? The study also asks a secondary comparative question: what happens when a general-purpose slow-adaptive algorithm such as EcoTrack is deliberately driven by the same fast power quality signal for which it was not designed?

This paper makes four contributions:

1. It formulates rare power-quality transient capture as a resource-minimization problem subject to a low-latency event-capture constraint, rather than treating sampling primarily as a smooth signal-estimation problem.
2. It proposes TASA, a two-state finite-state machine that combines low-rate baseline monitoring, a lightweight trigger detector, pre-trigger buffering, deterministic full rate capture, and hysteresis-based termination.
3. It implements TASA in C and evaluates it through STM32F103/Wokwi firmware simulation. In the supplied evaluation trace, TASA uses 588 bytes of static RAM, captures all 80 ground-truth event samples, and reduces the time-weighted average sampling rate by 72.7% relative to a 200-Hz fixed-rate reference.
4. It provides a faithful C re-implementation of EcoTrack and uses it as a deliberately cross-domain stress test, exposing a concrete failure mode: the algorithm's integer rate parameter reaches zero throughout the injected sag and remains zero for most of the evaluated trace.

BACKGROUND AND RELATED WORK

IEEE Std. 1159-2019 classifies voltage magnitude disturbances using ranges expressed in per-unit (pu). In particular, a voltage sag is a reduction in RMS voltage to approximately 0.1–0.9 pu of nominal, while a voltage swell is an increase to approximately 1.1–1.8 pu, with event durations extending from fractions of a cycle to longer intervals [1]. IEC 61000- 4-30 specifies measurement methods for voltage magnitude, dips and swells, interruptions, transient voltages, frequency, harmonics, interharmonics, and related PQ parameters [2]. The present work focuses on the sampling problem surrounding voltage-magnitude events rather than on complete PQ classification. The synthetic evaluation therefore uses one sag and one swell as representative event classes.

Adaptive sampling is widely used to reduce sensing activity when a monitored quantity changes slowly or when energy availability constrains acquisition. Shu et al. developed an adaptive sampling method for automated water-quality monitoring with an explicit data-quality objective [8]; related work has addressed adaptive sampling in sensor networks and energy-harvesting systems [9]-[11]. Surveys of adaptive sampling further show that rate adaptation is commonly used to trade data quality against resource utilization [10]. Giordano et al. proposed EcoTrack, an energy-aware adaptive sampling mechanism based on a finite-state machine and additive increase/multiplicative-decrease control for battery-powered IoT systems [13]. These approaches demonstrate that adaptive sampling can reduce resource use. However, their optimization objectives differ from the present problem. Here, the primary concern is not merely to maintain an acceptable approximation of a slowly varying signal. Instead, the system must preserve a rare, bounded-duration event whose absence can invalidate the monitoring objective.

Power-quality research frequently uses synthetic disturbances because controlled signal generation allows event type, magnitude, duration, and noise to be varied systematically. Machlev et al. developed an open-source power-quality disturbance generator specifically to support reproducible PQ disturbance experiments [15]. The present study similarly uses a labeled synthetic signal, but its purpose is not classification. The labels define ground truth for evaluating whether the sampling mechanism retains event samples.

Event-triggered and non-uniform sampling provide related alternatives in which samples are acquired according to signal activity rather than at a fixed rate [16] – [18]. The relevant gap is therefore not the absence of adaptive or event-driven sampling itself. Rather, it is the mismatch between continuously adaptive rate control and a monitoring objective in which a short power quality event has substantially higher value than an additional sample during stable operation. TASA addresses this mismatch by separating the system into two explicit operating modes. The baseline mode is optimized for resource conservation; the triggered mode is optimized for event preservation. The transition between the two is driven by a lightweight detector rather than by gradual rate adaptation.

PROBLEM FORMULATION

Let $v(t)$ denote the monitored RMS voltage envelope normalized to nominal voltage:

$$v(t) = \frac{V_{RMS}(t)}{V_{nom}} \quad (1)$$

During stable operation,

$$v(t) \approx 1 \quad (2)$$

apart from measurement noise. A transient event e_i begins at t_i and persists for a finite duration d_i. For the evaluated event classes, the synthetic signal contains a sag near 0.4 pu and a swell near 1.3 pu. These values lie inside the voltage-magnitude ranges associated with the corresponding IEEE 1159 event classes.

Let $r(t)$ be the instantaneous sampling rate. TASA permits two rates:

$$r(t) \in \{r_b, r_f\}, \quad r_f > r_b \quad (3)$$

The baseline rate r_b is used when the signal is stable. The full rate r_f is used after a trigger is detected.

The time-averaged sampling rate over an observation interval T is

$$\bar{r} = \frac{1}{T}\int_0^T r(t)dt \quad (4)$$

If p_{trig} denotes the fraction of active observation time spent in the triggered state, then

$$\bar{r} = r_b\left(1 - p_{trig}\right) + r_f p_{trig} \quad (5)$$

The corresponding reduction relative to a fixed-rate reference operating at r_f is

$$R = 1 - \frac{\bar{r}}{r_f} \quad (6)$$

The sampling policy should minimize $\bar{r}$ while preserving a high probability of retaining the event samples after detection.

A general target can be expressed as

$$P(capture(e_i)|\delta_i \leq \delta_{max}) \geq 1 - \epsilon, \quad (7)$$

where δ_i is the detector latency for event e_i and ϵ is an allowable miss probability.

This study does not claim a probabilistic guarantee over a population of real grid events. Instead, the expression defines the design objective, while the experiment reports empirical recall on the labeled synthetic test scenario.

PROPOSED ALGORITHM: TASA

A. Two-State Finite-State Machine

TASA contains two states:

1. BASELINE (B): sample at r_b, run the lightweight detector, and continuously maintain a pre-trigger ring buffer.
2. TRIGGERED (T): sample at r_f, preserve the event and its surrounding context, and remain in the high-rate state until hysteresis condition is satisfied or the capture limit is reached.

The key difference from continuous rate adaptation is that TASA does not progressively increase and decrease the sampling rate. The transition is explicit. This design is related conceptually to event-triggered and level-crossing sampling, but differs in using a bounded two-state logger with explicit pre-trigger storage [19]-[21].

B. Trigger Detector

The detector uses two inexpensive tests combined with logical OR.

The first is a slope test:

$$D_s(t) = |v(t) - v(t-1)| > \tau_{slope} \quad (8)$$

The second is an RMS-deviation test:

$$D_r(t) = \left|RMS_{rolling}(t) - RMS_{baseline}\right| > \tau_{rms} \quad (9)$$

The combined trigger is

$$D(t) = D_s(t) \vee D_r(t) \quad (10)$$

The evaluated thresholds are $\tau_{slope} = 40$ ADC counts and $\tau_{rms} = 60$ ADC counts on the 10-bit simulated ADC range.

The slope detector is intended to respond quickly to abrupt event onset. The RMS detector provides additional sensitivity to sustained deviations for which a single-sample slope condition may no longer remain active.

C. Buffer Management

A circular pre-trigger buffer of N_{pre} samples is maintained continuously during BASELINE operation. When a trigger fires, this buffer provides context preceding the detection instant. The evaluated configuration is

$$N_{pre} = 32, \qquad N_{post} = 200.$$

The high-rate capture state continues sampling at

$$r_f = 200Hz.$$

A hysteresis counter h is reset whenever the trigger remains active and incremented when the trigger is inactive. The state returns to BASELINE when

$$h \geq H, \tag{11}$$

with $H = 10$, subject to the configured capture-buffer limit.

D. ALGORITHM

Algorithm 1 Trigger-Adaptive Sampling Algorithm (TASA)

$S \leftarrow$ BASELINE
initialize circular buffer B_{pre}
initialize capture buffer B_{cap}
$h \leftarrow 0$
Loop
  $v \leftarrow$ read_ADC()
  if $S = BASELINE$ **then**
    sample using r_b
    Append v to B_{pre}
    if $D(v) = 1$ **then**
      $S \leftarrow TRIGGERED$
      copy/drain B_{pre} into B_{cap}
      $h \leftarrow 0$
    end if

Else
sample using r_f
append v to B_{cap}
if $D(v) = 1$ **then**
$h \leftarrow 0$
Else
$h \leftarrow \mathrm{h} + 1$
end if
if $h \geq H$ or capture limit reached **then**
flush B_{cap}
clear B_{cap}
$S \leftarrow \mathrm{BASELINE}$
end if
end if
end loop

IMPLEMENTATION

A. Simulated Platform

The intended deployment class is an STM32L4-class microcontroller. Because the study was conducted without physical hardware, the experimental implementation uses an STM32F103 target in Wokwi [22]. The substitution is explicitly treated as a limitation; the primary conclusions concern algorithmic behavior and compiler-reported static memory rather than physical power consumption. The simulated ADC produced 10-bit values in the tested environment. Consequently, the detector thresholds are reported in ADC counts for this simulated configuration and should not be interpreted as universal thresholds for physical hardware.

Table 1: TASA Evaluation Configuration

Parameter	Value
Baseline rate r_b	50 Hz
Full rate r_f	200 Hz
Pre-trigger buffer N_{pre}	32 samples
Post-trigger limit N_{post}	200 samples
Hysteresis H	10 samples
Slope threshold τ_{slope}	40 ADC counts

RMS threshold τ_{rms}	60 ADC counts
ADC resolution in simulation	10 bit

B. Firmware

The firmware was written in C using the Arduino framework without an RTOS. A timer-driven execution structure samples the ADC according to the active state, executes the trigger detector, and manages transitions between BASELINE and TRIGGERED operation. TASA and the EcoTrack comparison implementation were compiled under equivalent ARM Cortex-M compilation settings using:

```
-mcpu=cortex-m3 -mthumb -Os
```

for the memory-footprint comparison

C. Synthetic Test Signal

The test signal contains a stable nominal RMS-voltage level of approximately 1.0 pu with approximately 1% Gaussian noise. Two labeled events are injected:

1. a 0.4-pu voltage sag lasting 1.0 s;
2. a 1.3-pu voltage swell lasting 0.6 s.

   The generated reference signal contains 1000 source samples at 50 Hz. Synthetic PQ disturbance generation is consistent with established reproducibility practices for PQ research [12]. Thus, the two ground-truth event durations correspond to 50 and 30 source samples, respectively, for a total of 80 labeled event samples.

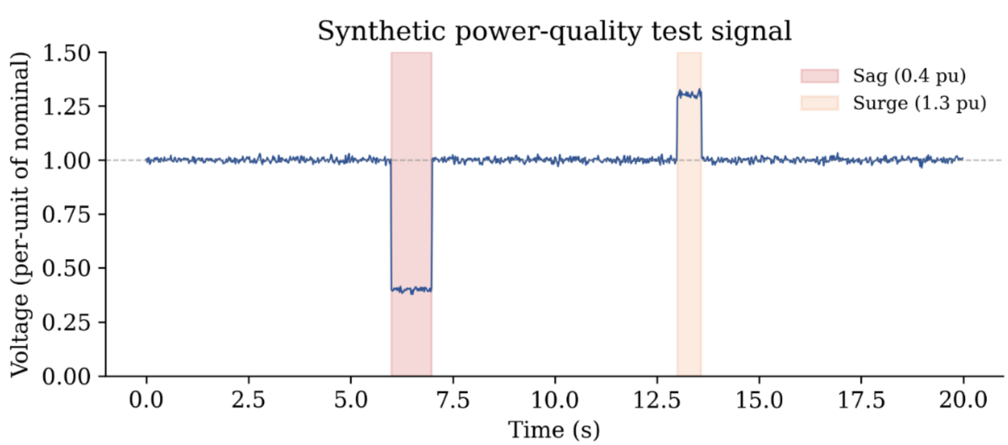

Fig 1:

EXPERIMENTAL EVALUATION

A. Baselines

Three sampling approaches were considered.

i. Fixed-Rate Sampling

The fixed-rate baseline continuously samples at

$$r_f = 200\ Hz. \tag{12}$$

This serves as the high-rate reference against which TASA's average sampling activity is compared.

ii. TASA

The proposed two-state trigger-adaptive strategy.

iii. EcoTrack

EcoTrack is published adaptive sampling approach based on an additive-increase and multiplicative-decrease mechanism [13]. It was re-implemented in C from the published algorithm description for the cross-domain comparison.

EcoTrack was designed for slowly varying resource signals rather than fast power-quality transients. Supplying the synthetic power-quality signal to the algorithm is therefore a deliberate stress test of the adaptive-sampling approach under a signal structure outside its original application domain. The results should not be interpreted as a general assessment of EcoTrack on its intended tasks.

B. Metrics

The evaluation reports:

i. Static RAM footprint

ii. Average sampling rate and reduction relative to the fixedrate reference

iii. Event-sample recall

iv. Event-sample precision

v. Trigger-state behavior

vi. Observed detection latency

For event-sample recall,

$$Recall = \frac{ground-truth\ event\ samples\ captured}{ground-truth\ event\ samples} \tag{13}$$

Precision us defined over the TASA trigger-selected sample set:

$$Precision = \frac{Triggered\ samples\ inside\ event\ windows}{all\ triggered\ samples} \qquad (14)$$

C. Memory Footprint

The compiler-verified static state footprints are shown in Table II. TASA requires 588 bytes, whereas EcoTrack requires 40 bytes. The ratio is

$$\frac{588}{40} = 14.70 \qquad (15)$$

Table 2: Static RAM Footprint

Algorithm	RAM (bytes)	Notes
TASA	588	Includes pre/post-trigger capture buffers and detector state
EcoTrack	40	No waveform capture buffer; tracks sampling-rate state

The larger memory requirement of TASA is primarily attributable to the bounded waveform capture buffers. EcoTrack does not store equivalent event waveform data, so the difference reflects the distinct objectives of the two algorithms.

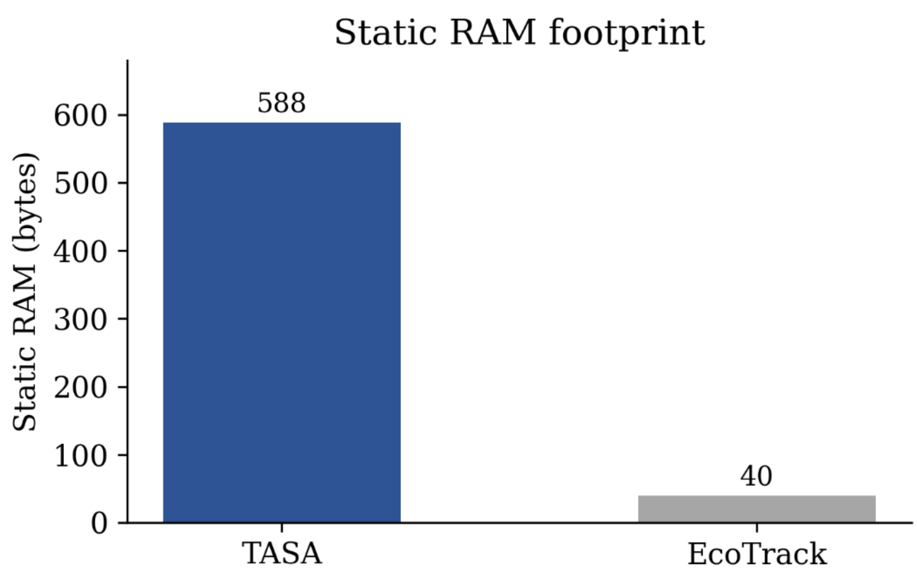

Fig. 2:

D. TASA Sampling Activity

The experiment reports 888 baseline samples and 112 triggered samples. The two triggered intervals are:

1. Event 1: Indices 301-366 inclusive, 66 triggered samples
2. Event 2: Indices 651-696 inclusive, 46 triggered sample.

Thus,

$$N_{triggered} = 66 + 46 = 112 \tag{16}$$

and

$$N_{baseline} = 888 \tag{17}$$

Using the configured rates, the corresponding active observation time are

$$T_b = \frac{888}{50} = 17.76\ s \tag{18}$$

$$T_t = \frac{112}{200} = 0.56\ s \tag{19}$$

The resulting active observation time is

$$T = T_b + T_t = 18.32\ s \tag{20}$$

The time-weighted average sampling rate is therefore

$$\bar{r}_{TASA} = \frac{888+112}{18.32} = 54.59\ Hz. \tag{21}$$

Relative to the 200Hz fixed-rate reference,

$$R = 1 - \frac{54.59}{200} = 72.7\% \tag{22}$$

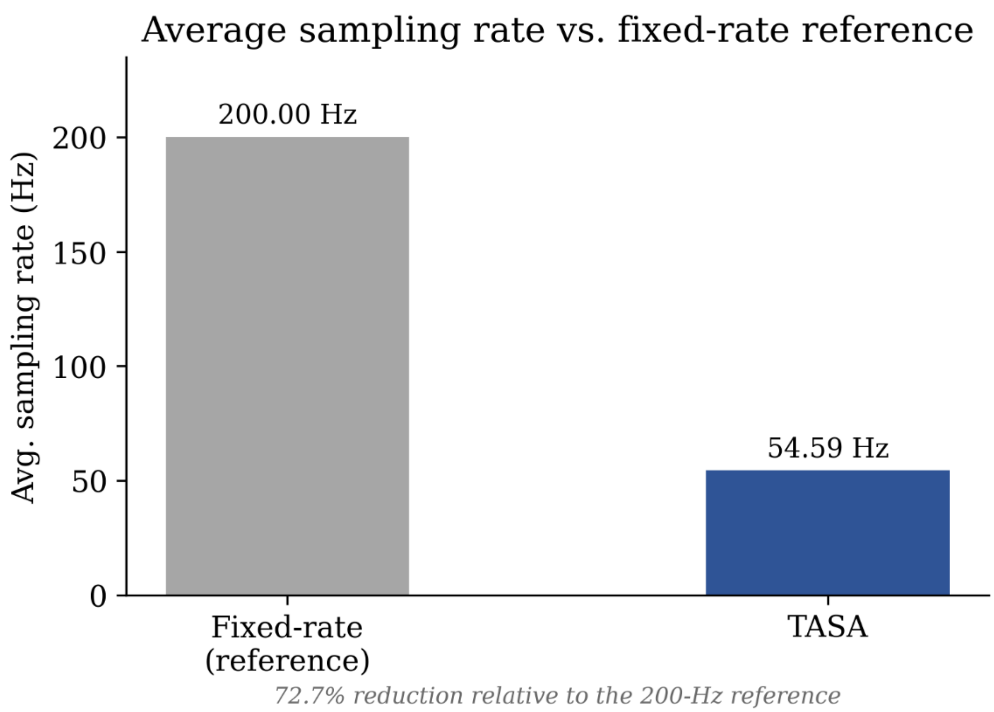

Fig. 3:

E. Transient Capture

The ground-truth event signal contains 80 event samples: 50 samples for the 1.0s sag and 30 samples for the 0.6s swell

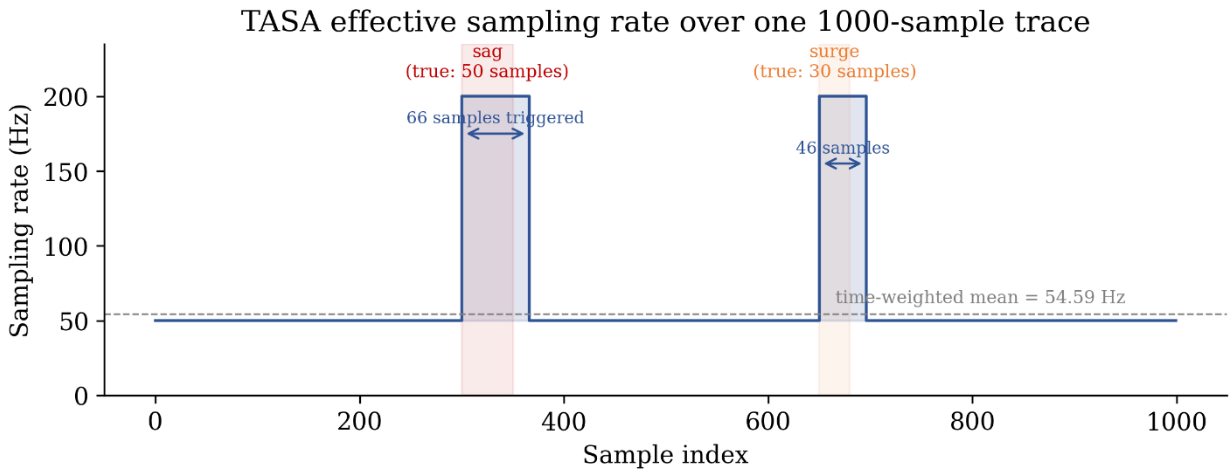

Fig 4:

The full TASA FSM captured all 80 event samples:

$$Recall_{TASA} = \frac{80}{80} = 100\% \tag{23}$$

$$Precision_{TASA} = \frac{80}{112} = 71.4\% \tag{24}$$

The lower precision is a consequence of deliberate capture margin and hysteresis. In this application, additional samples around an event provide context and protect against premature termination of the capture state.

Table 3: Transient Capture Results

Condition	Recall	Precision	Observation
TASA detector only	97.5%	-	Tail misses
TASA full FSM	100%	71.4%	Hysteresis
EcoTrack	0% during sag	-	k = 0 during sag

The detector-only result and full-FSM result should not be conflated. The detector can become non-triggering near the end of an event, while the hysteresis mechanism intentionally keeps the system in the triggered state long enough to preserve the event tail.

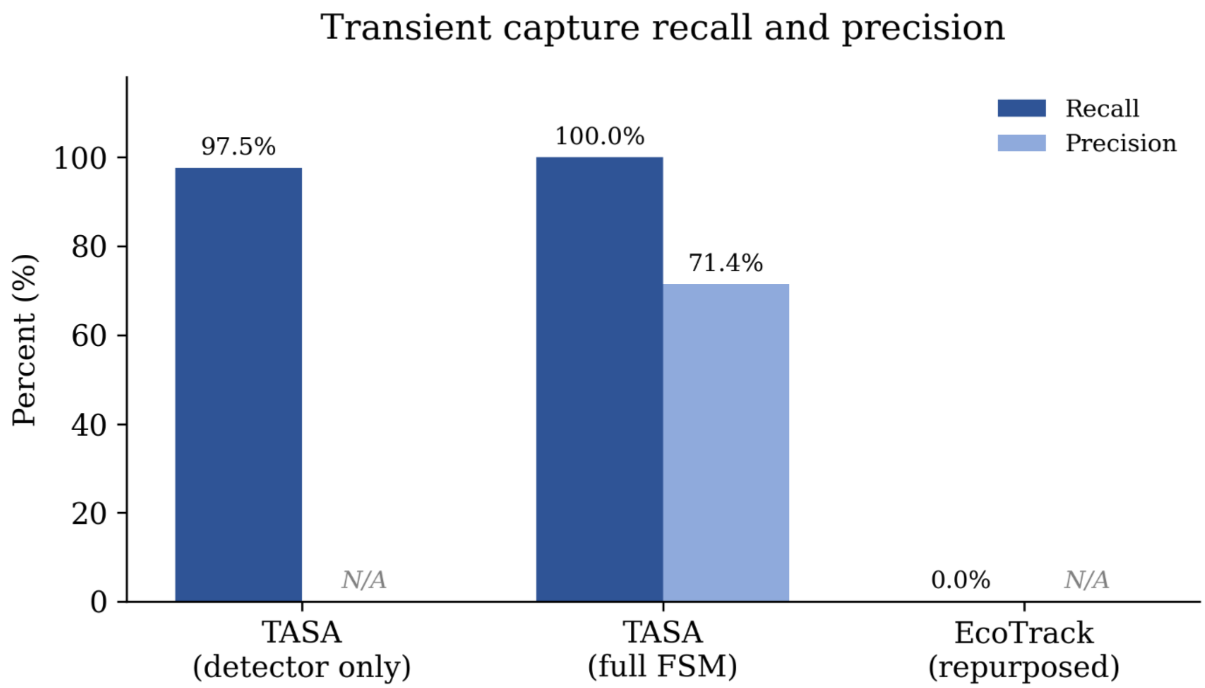

Fig 5:

F. Detection Latency

For the evaluated sag and swell events, the TASA trigger fired within one baseline sample period of the event onset. At

$$r_b = 50\ Hz, \tag{25}$$

One baseline period is

$$\delta_{max} = 20\ ms. \tag{26}$$

This is an observed result for the evaluated synthetic waveform and threshold configuration, not a universal upper bound for all real-world disturbances.

G. EcoTrack Cross-Domain Behavior

In thte evaluated sag scenario, the re-implemented EcoTrack adaptation logic responds to the negative excursion through multiplicative decrease. Its integer state variable k begins at 1, can be reduced by integer halving, and can increase additively up to a configured maximum of 200. Starting from the relevant low sampling rate state, integer halving causes the rate variable to collapse to zero, with no recovery within the evaluated event window.

Table 4: EcoTrack Trace Summary

Quantity	Observed Value
Static RAM	40 B
Configured k_{min}	0
Configured k_{max}	200
Observed k_{min}	0
Observed k_{max}	46
Ticks with $k = 0$	947/1000
Fraction with $k = 0$	94.7%
Sag k range	0-0
Swell k range	1-46

During the sag failure, k collapses to zero at the beginning of the sag and does not recover during that event. Consequently, no samples are acquired by the corresponding rate-control mechanism while the event is active. This produces 0% event capture for the sag in the evaluated cross-domain experiment. During the swell, the additive-increase mechanism raises k one unit at a time. Although k responds in the correct direction, its response is gradual relative to the short event duration. This illustrates a second mismatch between slow rate adaptation and rare transient capture.

This result is not presented as a flaw in EcoTrack's intended battery-resource application. Instead, it provides an example of why an adaptive mechanism whose assumptions are based on gradual signal evolution may be unsuitable for a rare, fast transient-capture tasks.

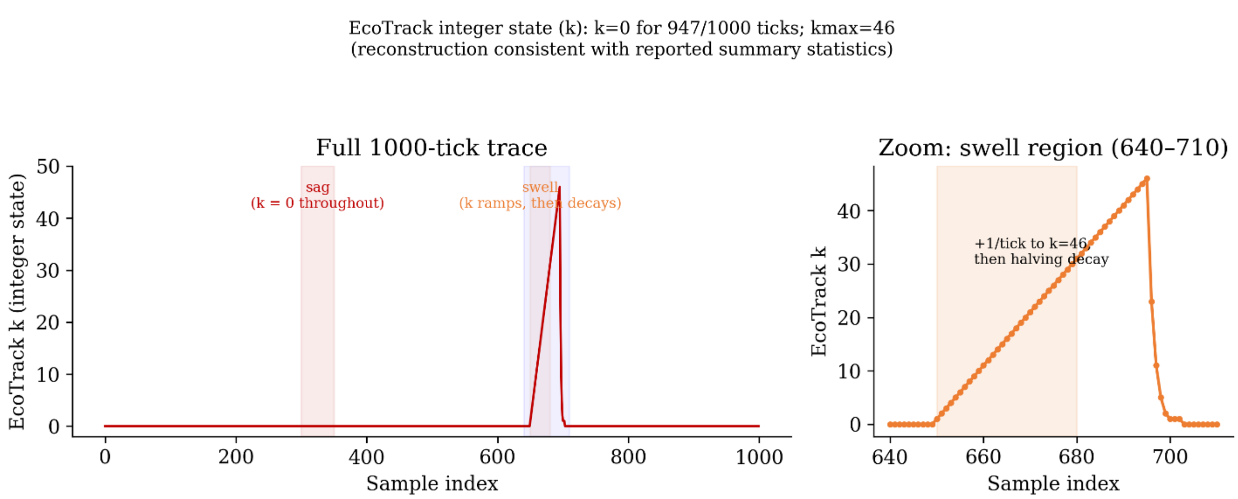

Fig 6:

H. Threshold Sensitivity

The slope threshold and RMS-deviation threshold were swept across a range of values using the same labeled synthetic signal and the full detection and hysteresis pipeline. The results indicated that recall was relatively robust across the tested slope threshold range from 20 to 100 ADC counts provided that the RMS threshold remained below the observed failure region. The RMS threshold was more sensitive: recall remained at 100% over a lower threshold region and degraded at substantially higher values. The operating point

$$\tau_{slope} = 40, \tag{27}$$

and

$$\tau_{rms} = 60. \tag{28}$$

was selected within the reported safe region rather than at the observed boundary at which recall degradation began.

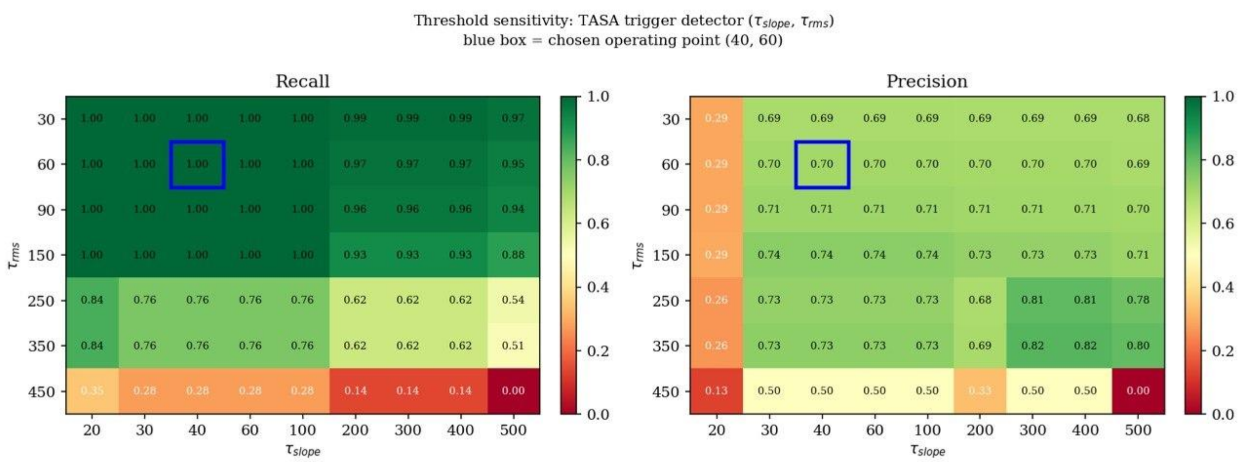

Fig 7:

DISCUSSION

A. Why the Two-State Design Matters

The main result is not simply that TASA uses fewer samples than a fixed-rate system. The more important result is the separation of resource-saving behavior from event-preservation behavior. During stable operation, TASA accepts that a lower sampling rate is sufficient and maintains a small amount of rolling context. Once a potential disturbance is detected, the algorithm does not attempt to optimize the rate continuously. It switches to a known full-rate mode. This design avoids a failure mode inherent to multiplicative rate control. AIMD itself is a well-established congestion control principle, originally developed for TCP congestion avoidance, where multiplicative decrease is intentionally used as a response to congestion [23]. A sudden negative excursion can be interpreted by an AIMD algorithm as a condition requiring reduced sampling, even though the same excursion may be exactly the event that the monitor must preserve.

B. Memory Tradeoff

TASA's 588-byte footprint is approximately 14.7 times the 40-byte EcoTrack footprint. This should not be interpreted as a direct efficiency ranking because the algorithms solve different storage problems. EcoTrack stores a small amount of state needed to control sampling activity. TASA stores actual waveform data so that a detected event can subsequently be inspected. The extra memory is therefore the price of turning a detection decision into an event record. For a microcontroller with tens of kilobytes of RAM, a sub-kilobyte algorithmic state footprint is modest in absolute terms, although the actual feasibility depends on the memory required by the ADC driver, communication stack, filesystem, and other application functions.

C. Average Sampling Activity

The time-weighted average sampling rate of 54.59 Hz, corresponding to a 72.7% reduction relative to 200 Hz. This number should be described as a reduction in sampling activity, not as a directly measured reduction in electrical power consumption. Actual energy savings depend on ADC acquisition cost, CPU execution, memory writes, storage operations, communication, and the power-management strategy of the final hardware.

D. Limitations

This work has several limitations:

1. Simulation-only validation: The results were obtained in Wokwi rather than on physical hardware. Real ADC noise, thermal drift, electromagnetic interference, analog-front-end behavior, and power-supply effects are therefore absent.
2. Target substitution: The intended deployment class is STM32L4, but the experiment uses STM32F103 because of simulator support. Platform-specific power conclusions are not claimed.
3. Synthetic data: The evaluation uses programmatically generated events rather than field-recorded grid disturbances.

4. Limited event diversity: Only one sag and one swell magnitude/duration configuration is used in the principal experiment. Interruptions, rapid voltage changes, harmonics, frequency excursions, and compound disturbances are outside the present scope.
5. Threshold generalizability: The detector thresholds were evaluated against the synthetic signal and its noise assumptions. Physical sensors may require recalibration.
6. Cross-domain EcoTrack comparison: EcoTrack is deliberately driven by a signal outside its intended battery state-of-charge application. The comparison is therefore a test of algorithm-class suitability, not a general benchmark of EcoTrack.
7. No direct power measurement: The study measures sampling activity and static memory, not physical energy consumption.

E. Threats to Validity

The principal threat to external validity is the difference between the simulated environment and a physical deployment. Real power systems may contain harmonic distortion, sensor noise, compound disturbances, and event combinations not represented by the evaluated synthetic signal.

CONCLUSION AND FUTURE WORK

This paper presented the Trigger-Adaptive Sampling Algorithm, a two-state sampling approach designed for rare power-quality transient capture on memory-constrained microcontrollers. TASA operates at a low baseline sampling rate during stable conditions and transitions to a higher capture rate when a lightweight detector identifies a potential disturbance. A bounded pre-trigger buffer preserves context preceding the detection instant, while hysteresis extends capture beyond the instantaneous detector response. In the evaluated firmware-simulation scenario, the full TASA finite-state machine achieved 100% recall across 80 labeled event samples, a 54.59 Hz time-weighted average sampling activity, required 588 bytes of static RAM, and reduced the average sampling rate by 72.7% relative to a 200-Hz fixed-rate baseline. A cross-domain comparison with a C re-implementation of EcoTrack demonstrated that a continuously adaptive rate mechanism designed for slowly varying signals can exhibit a failure mode when directly applied to the evaluated fast power-quality signal.

The results provide a simulation-based feasibility demonstration rather than a field-performance guarantee. Future work should therefore focus on three areas.

First, TASA should be deployed on physical STM32L4-class hardware with a real voltage-sensing front end and controlled disturbance injection. Second, the detector should be evaluated against public and field-recorded PQ datasets containing a broader range of disturbance types and noise conditions. Third, a complete system should address event storage, standardized event export, communications, and long-term energy management. COMTRADE and PQDIF provide established directions for subsequent event-data interchange [24], [25].

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